

Cell cycle-dependent cues regulate temporal patterning of the *Drosophila* central brain neural stem cells

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Gonzalo N Morales Chaya, Mubarak Hussain Syed 

Neural Diversity Lab, Department of Biology, University of New Mexico, Albuquerque, United States • Institute of Neuroscience, Howard Hughes Medical Institute, University of Oregon, Eugene, United States

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eLife Assessment

This **useful** manuscript reports findings indicating that cell cycle progression and cytokinesis both play a role in the transition of early to late neural stem cell fates. The imaging data are **solid** and mostly support the conclusions. However, experimental details are missing, the method of quantitation could be improved, and orthogonal approaches are needed to confirm the findings, which are based on loss-of-function approaches and are not sufficient to support some of the authors' conclusions. Lastly, there is no investigation of the underlying mechanism linking the cell cycle or cytokinesis to the changes (or lack thereof) of early and late NSC fates.

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Abstract

During nervous system development, diverse types of neurons and glia are sequentially generated by self-renewing neural stem cells (NSCs). Temporal changes in gene expression within NSCs are thought to regulate neural diversity; however, the mechanisms regulating the timing of these temporal gene transitions remain poorly understood. *Drosophila* type 2 NSCs, like human outer radial glia, divide to self-renew and generate intermediate neural progenitors, amplifying and diversifying the population of neurons innervating the central complex, a brain region crucial for sensorimotor coordination. Type 2 NSCs express over a dozen genes temporally, broadly classified as early and late-expressed genes. A conserved gene, *seven-up*, mediates early to late gene expression by activating ecdysone receptor (EcR) expression. However, the timing of EcR expression and, consequently, the transition from early to late gene expression remains unknown. This study investigates whether intrinsic mechanisms of cell cycle progression and cytokinesis are required to induce the NSC early-late transition. By generating mutant clones that arrest the NSC cell cycle or block cytokinesis, we show that both processes are necessary for the early-to-late transition. When NSCs are cell cycle or cytokinesis arrested, the early gene *Imp* fails to be downregulated and persists in the old NSCs, while the late factors *EcR* and *Syncrin* fail to be expressed. Furthermore, we demonstrate that the early factor *Seven-up* is insufficient to drive the transition, despite its normal expression in cell cycle- or cytokinesis-inhibited NSCs. These results suggest that both

cell-intrinsic (cell cycle/cytokinesis) and extrinsic (hormone) cues are required for the early-late NSC gene expression transition.

Introduction

From insects to humans, multipotent neural stem cells (NSCs) generate diverse neurons and glia, which mediate proper nervous system function. Both spatial and temporal patterning programs regulate the diversity of neurons generated during nervous system development (Doe, 2017 [DOI](#); El-Danaf et al., 2023 [DOI](#); Enriquez et al., 2015 [DOI](#); Erclik et al., 2017 [DOI](#); Hamid et al., 2023 [DOI](#); Kohwi and Doe, 2013 [DOI](#); Malin and Desplan, 2021 [DOI](#); Oberst et al., 2019 [DOI](#); Sen, 2023 [DOI](#); Skeath and Thor, 2003 [DOI](#)). While much is known about the regulation of spatial identity, the mechanisms underlying temporal identity are not entirely understood. Elucidating the processes through which NSCs undergo temporal patterning, enabling them to produce diverse neurons and glial cells in the nervous system, is critical for understanding brain development and disease. Studies on the *Drosophila* nervous system have identified temporal identity genes that express temporally in NSCs and diversify neural cell types (Bayraktar and Doe, 2013 [DOI](#); Eldred et al., 2018 [DOI](#); Isshiki et al., 2001 [DOI](#); Li et al., 2013 [DOI](#); Maurange et al., 2008 [DOI](#); Syed et al., 2017 [DOI](#); Yang et al., 2015 [DOI](#)). Similar temporal identity programs were later shown to regulate vertebrate retinal and cortical cell type specification (Alsiö et al., 2013 [DOI](#); Elliott et al., 2008 [DOI](#); Javed et al., 2023 [DOI](#); Kohwi and Doe, 2013 [DOI](#); Koo et al., 2023 [DOI](#); Pebworth et al., 2021 [DOI](#); Telley et al., 2019 [DOI](#)). In *Drosophila* embryonic NSCs, these gene transitions were thought to be intrinsically regulated; however, our recent findings in larval central brain NSCs identified the role of extrinsic steroid growth hormone ecdysone in the early-to-late gene transition (Syed et al., 2017 [DOI](#)). Recently, thyroid hormone signaling has been identified as regulating the early-to-late born cone cell type in human retinal organoid cultures (Eldred et al., 2018 [DOI](#)). Over the past few years, various temporally expressed genes have been identified in both invertebrates and vertebrates; however, the factors that determine the precise timing of these gene transitions are poorly understood.

Based on division patterns, the *Drosophila* NSCs can be classified as type 0, type 1, or type 2 NSCs. Type 0 NSCs generate a single post-mitotic neuron with each division and have only been observed in embryonic stages (Arefin et al., 2020 [DOI](#); Baumgardt et al., 2014 [DOI](#)). Type 1 NSCs generate a ganglion mother cell (GMC) with each division; each GMC makes two post-mitotic neurons (Doe, 2017 [DOI](#); Doe and Goodman, 1985 [DOI](#); Rossi and Desplan, 2017 [DOI](#)). While type 1 NSCs comprise the majority of neural stem cells in the brain, type 2 NSCs—though limited to just 16—generate more complex and diverse progeny by exhibiting a more complex division pattern, producing approximately 5,000 neurons of at least about 160 distinct cell types (Epiney et al., 2025 [DOI](#)). Type 2 NSCs divide to self-renew and generate Intermediate Neural Progenitors (INPs); each INP undergoes a series of 4-6 self-renewing divisions to produce 4-6 GMCs, which each generate two neurons (Bello et al., 2008 [DOI](#); Boone and Doe, 2008 [DOI](#); Bowman et al., 2008 [DOI](#); Wang et al., 2014 [DOI](#)). Recent temporal RNA-sequencing experiments have revealed that type 2 NSCs undergo a gene expression cascade involving transcription factors (TFs) and RNA-binding proteins (RBPs). In early larvae 0-60h after larval hatching (ALH), the NSCs express TFs, Seven-up (Svp), Castor (Cas), Chinmo, and RBPs, IGF-II mRNA-binding protein (Imp), and Lin-28. In late larvae (60-120h ALH), the NSCs express TFs, Ecdysone receptor (EcR), Ecdysone-induced protein 93F (E93), Broad and RBP, Syncrin (Syp) (Ren et al., 2017 [DOI](#); Syed et al., 2017 [DOI](#)). The early-expressed conserved transcription factor Svp regulates the expression of EcR, thereby rendering NSCs competent to respond to the extrinsic hormonal signal, which triggers the early-late gene expression switch (Syed et al., 2017 [DOI](#)). While the role of the extrinsic factor Ecdysone in triggering the early-late transition is understood, nothing is known about how intrinsic factors drive this transition and what regulates the timing of these events.

In vertebrate cortical development, the cell cycle plays a key role in regulating a neural progenitor's ability to respond to extrinsic cues. Transplanted progenitors in a late-stage host exhibit different responses to these cues depending on their cell cycle stage at the time of transplantation (McConnell and Kaznowski, 1991 [↗](#); Ohnuma and Harris, 2003 [↗](#)). In *Drosophila*, several studies have demonstrated a relationship between the cell cycle and temporal gene expression programs in NSCs. Interestingly, this relationship differs markedly between embryonic and larval stages. In embryonic NSCs, the classic temporal cascade—Hunchback > Kruppel > Pdm1/2 > Castor—is largely cell-intrinsic and can proceed in isolated, G2-arrested cells (Grosskortenhaus et al., 2005 [↗](#)). While the transition from Hunchback to Kruppel requires cytokinesis, subsequent transitions (Kruppel to Pdm1/2 to Castor) occur even when cell cycle progression and cytokinesis are blocked.

In contrast, larval NSCs exhibit temporal transitions from early factors (Imp/Chinmo) to late factors (Syp/Broad) that depend on the cell's metabolic state and progression through the G1/S phase of the cell cycle (van den Aamele and Brand, 2019 [↗](#)). Notably, at the end of larval neurogenesis, late temporal programs schedule the end of neurogenesis, and NSCs undergo cell cycle exit or apoptosis. These late-stage transitions are governed by the Svp-mediated induction of the EcR and its downstream effector E93 (Pahl et al., 2019 [↗](#); Homem et al., 2014 [↗](#); Maurange et al., 2008 [↗](#); Syed et al., 2017 [↗](#)). However, whether Svp and EcR expression—and thus the timing of early-to-late transitions—are dependent on cell cycle progression remains unknown. Furthermore, it is not yet clear whether similar temporal and cell cycle interactions occur in type I and type II NSCs. Moreover, it is unclear whether the timing of NSC gene expression is driven by organismal cues (e.g., hormones or signaling pathways), internal cues (e.g., a cell cycle counting clock), or both. Here, we utilize larval central brain NSCs to investigate whether cell cycle and cytokinesis regulate the timing of the temporal gene expression program.

Results

Cell cycle and cytokinesis are both required for the early-late temporal factor switch in type 2 NSCs

Our previous work revealed that in type 2 NSCs, the transition from early to late temporal factors occurs around 55h ALH (Syed et al., 2017 [↗](#)). To better understand the mechanisms regulating the precise timing of these gene expression shifts, we next focused on the roles of the cell cycle and cytokinesis. While the cell cycle has been shown to influence temporal patterning in some systems, it appears to be dispensable in others, suggesting context-dependent roles in regulating these transitions (van den Aamele and Brand, 2019 [↗](#); Grosskortenhaus et al., 2005 [↗](#); Syed et al., 2017 [↗](#)). To investigate whether the temporal progression of type 2 NSCs requires cell cycle progression, we blocked the NSC cell cycle or cytokinesis in all type 2 NSCs starting at 0h ALH (the experimental model is shown in **Figure 1** [↗](#)). We blocked the cell cycle by knocking down Cyclin-dependent kinase 1 (Cdk1) at 0h ALH and assayed the temporally expressed early and late genes at 72h ALH. The Cdk1RNAi effectively blocked the cell cycle, as we could not observe any GFP-labeled progeny near the NSCs, and the NSC cell volume was larger than that of the control, which has also been attributed to normal cell-cycle progression (Hartenstein et al., 1987 [↗](#)) (**Figure S1** [↗](#)). While all control type 2 NSC lineages showed normal temporal factor expression progression, Cdk1RNAi type 2 NSCs exhibited prolonged expression of the early factor Imp into later stages of development, and failed to upregulate the late factors Syp and EcR (**Figure S2** [↗](#)), confirming that cell cycle is essential for the early-to-late gene transition.

To further ensure the cell-specificity of this phenotype, we used the flip-out approach and induced random single cell clones of type 2 NSCs. We induced random clones by giving a heat shock at 0h ALH using the genotype *hsflp*, *10xUAS-mCD8G-FP*; *act>FRT-stop-FRT>gal4*. Both control RNAi (UAS-mCherryRNAi) and experimental (UAS-Cd-k1RNAi) clones were labeled by GFP, and type 2 hits

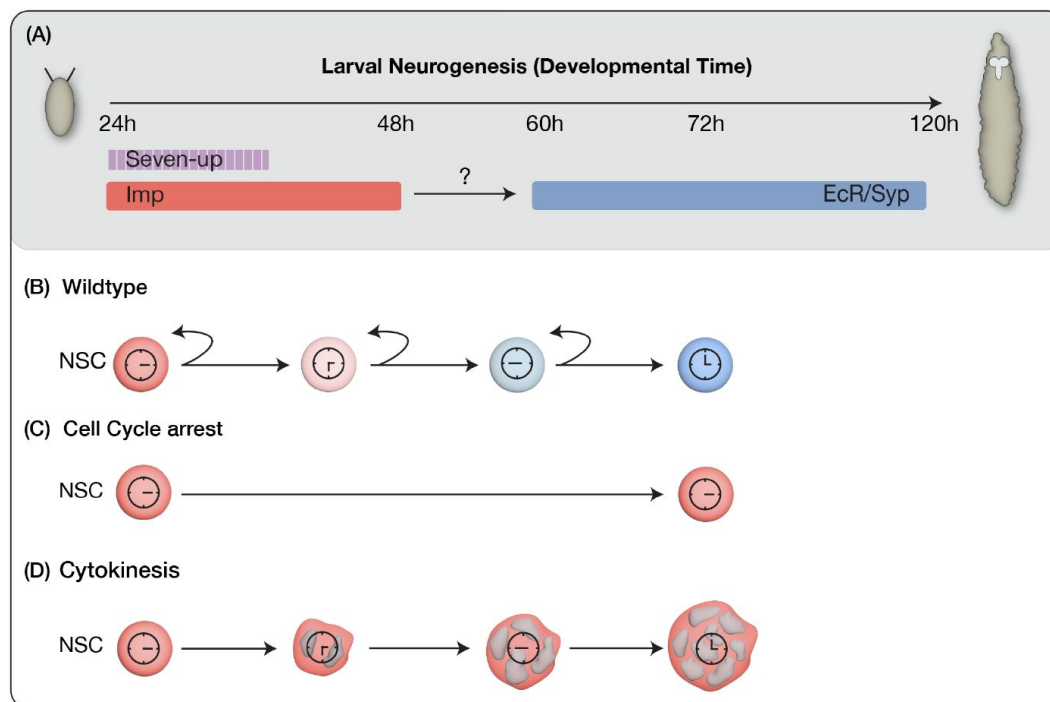


Figure 1

Models for early-to-late temporal progression transition of type 2 NSCs.

(A) Depicts timing of NSC temporal factor expression in type 2 NSCs during *Drosophila* larval development (B) Wild type NSCs undergo normal early to late temporal factor progression (C) Cell cycle progression as a regulator to allow timely expression of factors in NSCs (D) Cytokinesis dependent regulation of temporal gene expression.

were confirmed by lack of the type 1 Asense marker. After heat shock, we assayed the clones in 72h ALH larvae. In the control type 2 NSCs, as anticipated at 72h ALH, minimal or no expression of the early factor Imp was observed, alongside high expression of late factors EcR and Svp (**Figure 2A-2C**; [quantified in Figure 2J](#)). In contrast, the experimental type 2 NSCs, where the cell cycle was blocked, exhibited persistent expression of the early factor Imp and no expression of late factors EcR and Svp at 72h ALH (**Figure 2D-2F**; [quantified in Figure 2J](#)), thus affirming that cell cycle progression is indispensable for the early-late switch in temporal factor gene expression.

The failure to switch from early to late factors could arise from blocking nuclear divisions (as part of a gene expression “clock” or “timer”) or from preventing cytokinesis, resulting in a daughter cell that may deliver feedback signals or partition transcriptional regulators out of the NSC. To discern between these possibilities, we inhibited NSC cytokinesis, halting NSC progeny production while leaving nuclear divisions unaffected ([Grosskortenhaus et al., 2005](#)). To do this, we induced clones using pavarotti RNAi (UAS-pavR-NAi), a known inhibitor of cytokinesis that, when knocked down, produces large cells with multiple nuclei ([Adams et al., 1998](#)). Similar to our previous observations (**Figure S2**), control type 2 NSCs showed no Imp expression and normal EcR/Svp expression (**Figure 2A-2C**; [quantified in 2J](#)), while the experimental pavRNAi type 2 NSCs resulted in large multi-nucleated NSCs (**Figure S1 C-C'**) that maintained Imp expression and lacked EcR/ Svp expression (**Figure 2 G-2I**; [quantified in 2J](#)). We conclude that both nuclear division and cytokinesis are essential for the early-late gene expression switch in type 2 NSCs (**Figure 2K**).

The switching factor Svp is normally expressed in cell-cycle arrested NSCs

A conserved nuclear hormone receptor, Svp, mediates the early-to-late gene switch by regulating EcR expression ([Dillon et al., 2024](#), [Maurange et al., 2008](#), [Syed et al., 2024](#)). The *svp* mutant type 2 NSCs stay in their early gene expression module and fail to exit the cell cycle at the end of larval life. The mutant type 2 NSCs fail to downregulate the expression of early factors Chinmo and Imp and show no expression of the late factors EcR, Broad, Syncrip, and E93. While Svp shows its peak expression at 18-24h ALH, EcR expression starts at around 55h ALH ([Dillon et al., 2024](#), [Ren et al. 2017](#), [Syed et al., 2017](#)). Although *svp* is required for EcR expression, what regulates the precise timing of EcR expression around 55h ALH is unknown. The lack of EcR expression in cell cycle-arrested Type 2 NSCs could be due to either defective Svp expression or the lack of a cell cycle counter or both.

To test whether cell-cycle-arrested NSCs fail to express Svp and thus produce a phenotype similar to the *svp* mutant, we assayed Svp expression in type 2 cell-cycle-arrested NSCs. Since Svp expression is transient in these NSCs, we used a *sv-pLacZ* line, similar to our previous work ([Syed et al., 2017](#)), to achieve more stable protein expression. Upon counting LacZ-positive Type 2 NSCs at 48h ALH, nearly all control type 2 NSCs showed *svpLacZ* expression (**Figure 3A**), and cell-cycle-arrested animals displayed similar expression levels with no statistically significant difference from controls (**Figure 3B**, [quantified in 3C](#)). We conclude that the timing of Svp expression is normal in cell-cycle-arrested NSCs.

Svp and cell cycle progression are independently required for the early-late temporal factor switch

Since Svp expression was normal in cell-cycle arrested Type 2 NSCs, we wondered whether the cell cycle acts as a timer for the precise EcR expression following Svp expression. To address this question, we generated type 2 NSC clones expressing either Cdk1RNAi or pavRNAi at 42h ALH (see methods) well after the Svp expression window (18-24h ALH). We assayed for expression of NSC

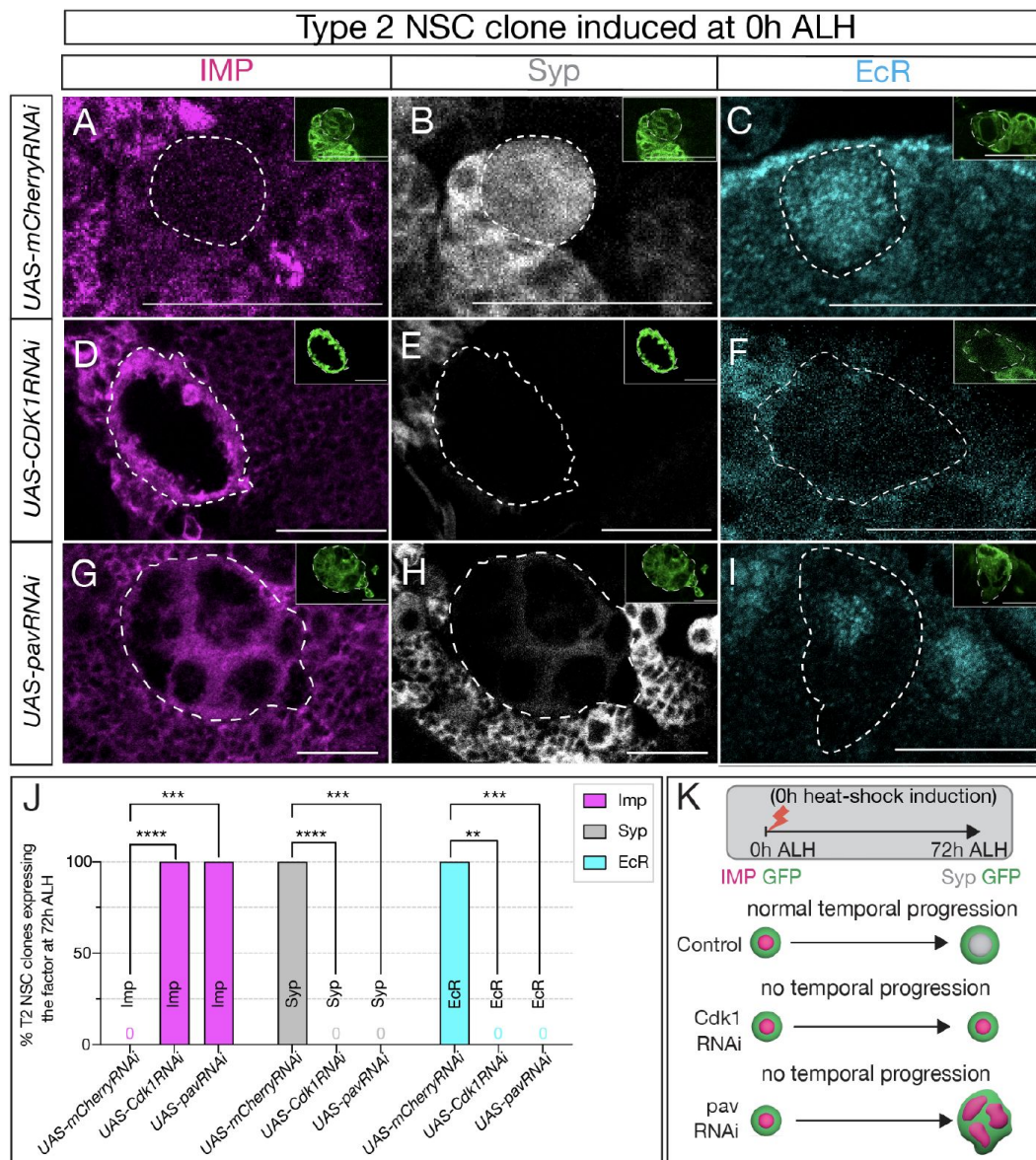


Figure 2

Cell cycle and cytokinesis are required for temporal gene expression progression in type 2 NSCs.

(A-C) Type 2 NSC control mCherryRNAi clones (circled) at 72h ALH show normal temporal gene expression progression. Early factor Imp is off, and late factors Syp and EcR are on at 72h ALH (D-F) Cdk1RNAi type 2 NSC clone fail to down-regulate early factor Imp, and late factors Syp and EcR are not turned on. (G-I) Similarly, cytokinesis-blocked type 2NSC clones using PavRNAi fail to downregulate Imp and upregulate Syp and EcR. (J) Quantification of early (Imp) and late (EcR, Syp) temporal factor expression in mCherryRNAi, Cdk1RNAi, and PavarottiRNAi type 2 NSCs ($n \geq 4$ clones per genotype). Statistical analysis by Fisher's exact test: Imp: 0/5 vs 11/11 clones in Cdk1RNAi ($p = 0.0002$), 0/5 vs 8/8 in pavarottiRNAi ($p = 0.0008$); EcR: 6/6 vs 0/4 in Cdk1RNAi ($p = 0.0048$), 6/6 vs 0/7 in pavarottiRNAi ($p = 0.0006$); Syp: 5/5 vs 0/11 in Cdk1RNAi ($p = 0.0002$), 5/5 vs 0/8 in pavarottiRNAi ($p = 0.0008$). Only one clone was found per animal. (K) Representation of experimental layout for inducing clones at 0h ALH and analysis at 72h ALH. Type 2 NSCs are identified as Asense-negative large cells expressing UAS-mcd8::GFP (green insets). Scale bars represent 20 μ m.

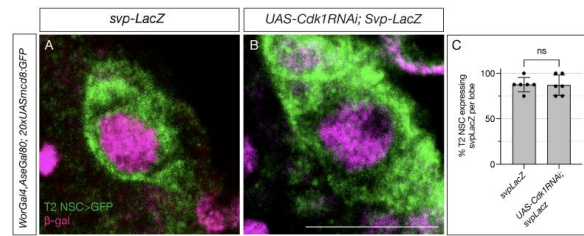


Figure 3

Switching factor Svp expresses normally in cell-cycle arrested type II NSCs.

(A) Control type 2 NSCs marked in green show Svp-LacZ expression at 48h ALH. (B) Cell-cycle-arrested (Cdk1RNAi) type 2 NSCs express Svp-LacZ similar to the control (C) Quantification of LacZ expression in control and cell cycle blocked type 2 NSCs, n=6 for each genotype. Unpaired t-test, p-value = 0.9168. Scale bars represent 10um.

temporal markers at 72h ALH. If the transition fails to occur, it will confirm that Svp primes EcR expression, but the intrinsic cell cycle clock determines the precise timing. Conversely, if the transition occurs normally, cell cycle/cytokinesis plays a role in early gene expression (before 42h), but not in later, post-Svp expression. Compared to control type 2 NSC clones that showed normal Imp downregulation and EcR/Syp expression at 72h ALH (**Figure 4A-4C**; **quantified in 4J**), Cdk1 and pav mutant type 2 NSC clones had persistent Imp expression and no EcR/Syp expression (**Figure 4D,4I**; **quantified in 4J**), confirming that cell cycle progression post-Svp expression is critical for precise timing of EcR and early to late gene transition (**Figure 3K**).

Cell cycle and cytokinesis are required for the early-late temporal factor switch in type 1 NSCs

So far, we have focused on type 2 NSCs, which are relatively few; however, the majority of larval NSCs are type 1 NSCs, which express the same early and late temporal factors as type 2 NSCs (Syed et al., 2017). The timing of EcR expression and the transition from early to late gene expression also matches the type 2 NSC pattern (Syed et al., 2017). Thus, we sought to determine if cell cycle and cytokinesis also regulate temporal gene transitions in type 1 NSCs.

Similar to our previous experiments, we generated clones using the genotype hsFLP, UAS-mCD8GFP; act>FRT-stop-FRT>gal4 / UAS-Cdk1RNAi or UAS-mCherryRNAi and assayed GFP and Asense positive type 1 NSCs. Control RNAi induced at 0h and assayed at 72h showed normal down-regulation of Imp and expression of the late factors Syp (**Figure 5A-C** and **5K-M**; **quantified in 5J and 5T**). In contrast, Cdk1 or pavRNAi type 1 NSC clones induced at 0h and assayed at 72h maintained Imp expression and lacked EcR/Syp expression (**Figure 5D-I**; **quantified in 5J**).

Similarly, Cdk1 or pavRNAi induced at 42h -- after the Svp expression window -- and assayed at 72h also maintained Imp expression and lacked EcR/ Syp expression (**Figure 5N-S**; **quantified in 5T**). This suggests that most central brain neural stem cells possess a cell-cycle and cytokinesis-dependent, cell-intrinsic mechanism that regulates the Imp/Syp expression transition. Our studies indicate that all central brain NSCs require cell cycle and cytokinesis, together with the switching factor Svp expression, to drive the early-late temporal factor expression transition.

Discussion

Temporal patterning plays a crucial role in specifying the diverse cell types of the nervous system. While many temporally expressed genes have been identified in both Drosophila and vertebrate neural progenitors (Javed et al., 2023; Koo et al., 2023; Pebworth et al., 2021; Syed et al., 2017; Telley et al., 2019; Walsh and Doe, 2017), what governs the timing of these gene expression transitions is not clearly understood. Is there an intrinsic timer within NSCs that counts and regulates the precise timing of temporal gene transitions? Is such a mechanism linked to cell-cycle progression and cytokinesis? How are NSC intrinsic and extrinsic environmental cues coupled? This study investigated the cell-intrinsic mechanisms regulating the early-to-late temporal gene expression transitions. Using Drosophila larval central brain NSCs, we identified that both cell cycle and cytokinesis-dependent cell-intrinsic timers regulate the progression of gene expression. In cytokinesis-blocked NSCs, the transition from early Imp expression to late EcR and Syp expression was impeded in both type 1 and type 2 NSCs. Upon temporal cell cycle block at 42h ALH, we showed that normal cell cycle progression is still essential for EcR expression despite the Svp expression being normal in the NSCs. These findings provide insights into the possible mechanisms regulating the temporal gene transition timing and also reveal the complex interplay of the cell cycle, cell-intrinsic genes, and cell-extrinsic hormonal cues in determining the precise timing of neuron type production within neural progenitors.

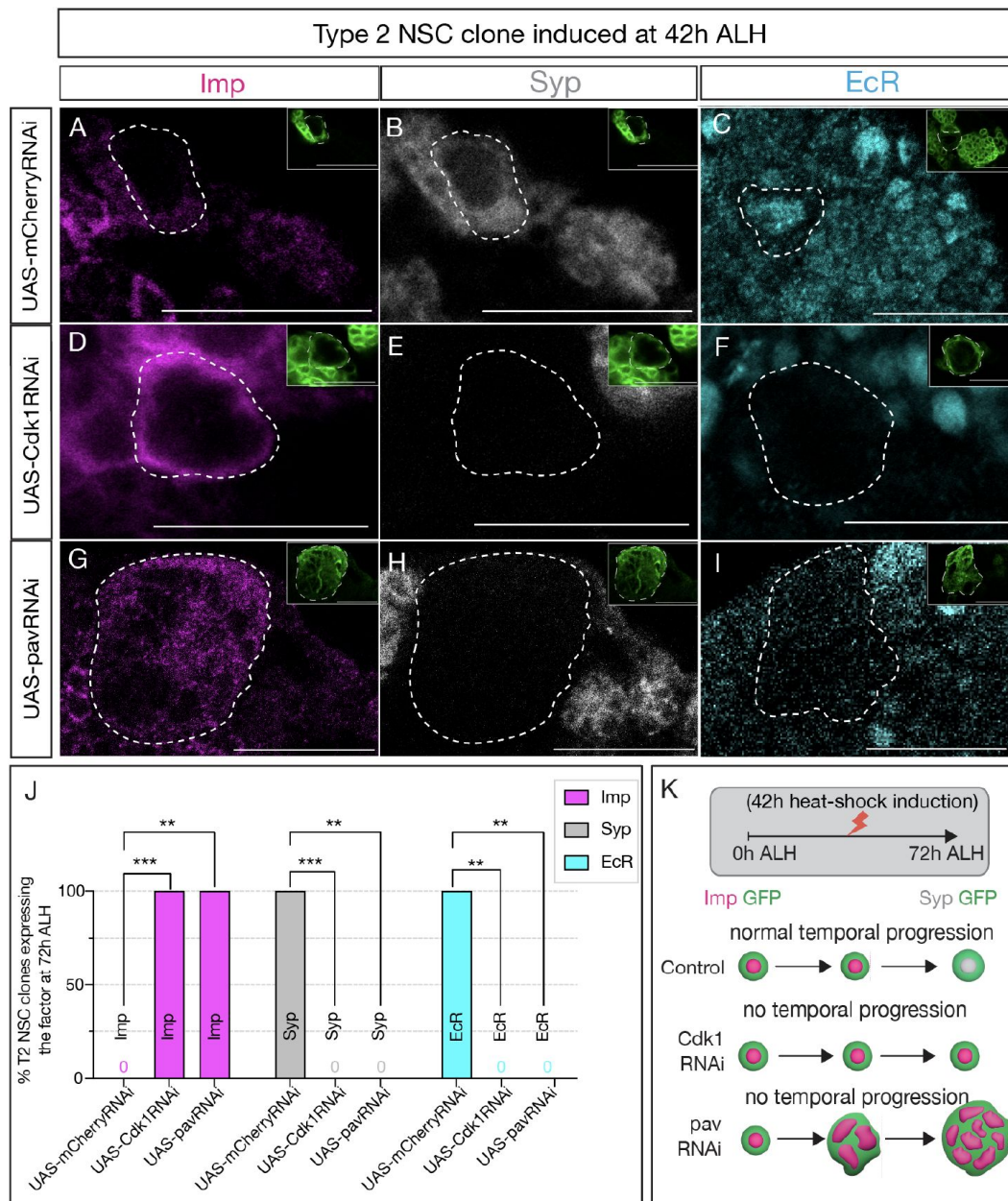


Figure 4

Early Svp expression is not sufficient to drive temporal factor progression in type 2 NSCs.

(A-C) Type 2 NSC control mCherryRNAi clones (circled) induced at 42h ALH and stained at 72h ALH show normal progression of temporal factors. Early factor Imp is off, and late factors Syp and EcR are on. (D-F) Cdk1RNAi clones show consistent failure to down-regulate early factor Imp and activate Syp and EcR (G-I) Likewise, pavRNAi clones fail to express EcR and Syp and consistently express early factor Imp. (J) Quantification of early (Imp) and late (EcR, Syp) temporal factor expression in control (UAS-mCherryRNAi), UAS-Cdk1RNAi, and UAS-pavarottiRNAi type 2 NSCs ($n \geq 4$ clones per geno-type). Statistical analysis by Fisher's exact test: Imp: Cdk1RNAi vs control $p = 0.0002$, pavRNAi vs control $p = 0.0079$; EcR: Cdk1RNAi vs control $p = 0.0079$, pavRNAi vs control $p = 0.0079$; Syp: Cdk1RNAi vs control $p = 0.0002$, pavRNAi vs control $p = 0.0079$. Only one clone was found per animal. (K) Representation of experimental layout. Clones are identified as Asense negative large cells expressing UAS-mcd8::GFP (green insets). Scale bars represent 20um.

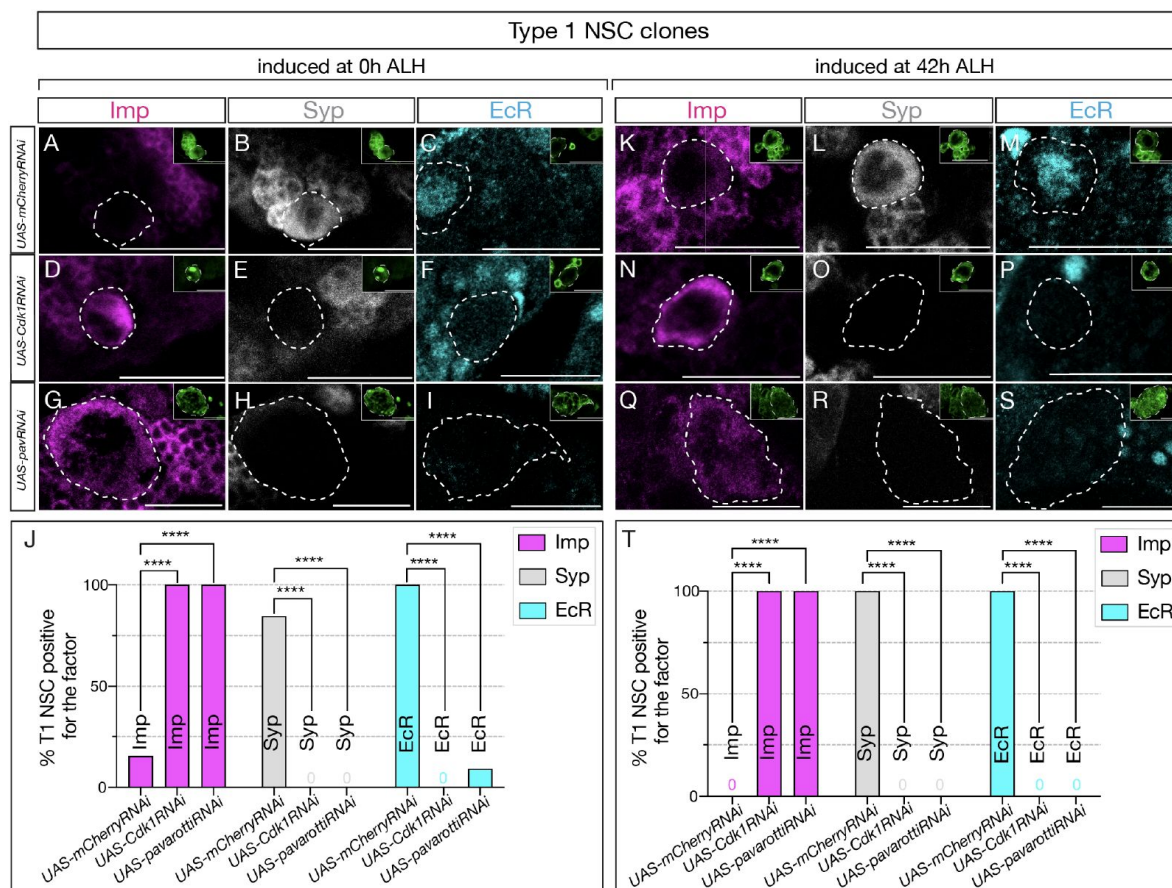


Figure 5

Cell cycle and cytokinesis are required for temporal progression in type 1 NSCs.

(A-I). Clones were induced at 0h ALH, and for (K-S), clones were induced at 42h ALH. (A-C) Type 1 NSC control RNAi clones show normal expression of temporal factor progression. (D-F) Cell cycle blocked type 1 NSC clones using Cdk1RNAi show failure to up-regulate EcR and Syp, and consistent expression of early factor Imp. (G-I) pavRNAi type I NSC clones fail to downregulate early factor Imp, and late factors Syp and EcR fail to express. (K-M) Control type 1 NSC clones induced at 42h ALH show normal expression of early and late factors, while Cdk1RNAi clones (N-P) and pavRNAi clones (Q-S) fail to downregulate Imp and fail to activate late factors Syp and EcR. (J, T) Quantification of early and late factor expression of 0h and 42h ALH-induced clones. (J) In clones induced at 0h ALH, Imp was expressed in 10/10 Cdk1RNAi clones (2/13 control, $p = 0.000067$) and 12/12 pavRNAi clones (2/13 control, $p = 0.000020$). EcR was not detected in Cdk1RNAi (0/11) or pavRNAi clones (0/11), while it was expressed in 12/12 control clones ($p = 0.000001$ for both). Syp was absent in Cdk1RNAi (0/10) and pavRNAi clones (0/12), while present in 11/13 control clones ($p = 0.000067$ and $p = 0.000020$, respectively). (T) In clones induced at 42h ALH, Imp expression persisted in 10/10 Cdk1RNAi (0/10 control, $p = 0.000011$) and 13/13 pavRNAi clones (0/10 control, $p = 0.000001$). EcR expression was not detected in Cdk1RNAi (0/8) or pavRNAi clones (0/13), compared to 10/10 control ($p = 0.000023$ and $p = 0.000001$). Syp expression was not detected in Cdk1RNAi (0/10) or pavRNAi clones (0/13), but was regular in 10/10 control ($p = 0.000011$ and $p = 0.000001$). Only one clone was found per animal. For each genotype $n \geq 8$. Type 1 clones are identified as Asense-positive, large cells expressing mcd8::GFP (green insets). Scale bars represent 20 μm .

The possible role of cell cycle and cytokinesis in regulating temporal patterning and neuronal identity has been studied in vertebrates and invertebrates (Doe, 2008 [↗](#); Grosskortenhaus et al., 2005 [↗](#); Hagey et al., 2020 [↗](#); Kawaguchi, 2019 [↗](#); van den Aamele J, Brand AH. 2019). In vertebrates, the G2/M phase is required for the sequential generation of layer-specific cortical neurons. Knockdown of G2/M regulators has been shown to prevent radial glial cells from maintaining a “young,” multipotent state, leading to premature differentiation and a bias toward the production of early-born neurons (Hagey et al., 2020 [↗](#); Kawaguchi, 2019 [↗](#)). In the case of *Drosophila* embryonic NSCs, which sequentially express Hunchback, Kruppel, Pdm1, and Castor, cell cycle is dispensable for these transitions. The initial Hunchback to Kruppel expression requires cytokinesis, while the Kruppel to Pdm1 to Castor transition occurs normally in G2 arrested NSCs (Grosskortenhaus et al., 2005 [↗](#)). In contrast, our studies on larval NSCs show that both cell cycle and cytokinesis are essential for the early Imp/Chinmo to late EcR/ Syncrip gene transition. The temporal gene progressions of embryonic NSCs occur normally in isolated NSCs, suggesting that these gene transitions are regulated intrinsically. In contrast, larval NSCs express EcR and require extrinsic growth hormone, ecdysone, for the transition from early to late gene expression. Our findings suggest that NSCs with more complex division patterns and dividing in situations where the growth and division patterns need to be coordinated require both cell-intrinsic and extrinsic cues. Studies on larval optic lobe NSCs thus far suggest that their temporal gene expression cascade is regulated cell-intrinsically, akin to embryonic NSCs (Konstantinides et al., 2022 [↗](#); Li et al., 2013 [↗](#)). However, it remains unclear whether these temporal transitions depend on progression through the cell cycle/cytokinesis or both. This raises the intriguing question of whether optic lobe NSCs follow an embryonic-like model, in which temporal patterning occurs independently of cell division, or resemble larval central brain NSCs, where temporal progression is tightly coupled to the cell cycle and cytokinesis.

Through the cell cycle, neural progenitors generate neurons populating distinct layers in the vertebrate cortex (Borrell and Calegari, 2014 [↗](#); Hagey et al., 2020 [↗](#); Ohtsuka and Kageyama, 2019 [↗](#); Salomoni and Calegari, 2010 [↗](#)); however, the relationship between cell cycle and temporal patterning has remained elusive. Given that cell cycle is essential for temporal gene transitions, it could be one potential way of timing the generation of distinct progeny with precise temporal control. While we were able to demonstrate the cell-intrinsic roles of cell cycle in timing temporal gene transitions, given the fact that NSC resides in a glial niche and in close proximity to its lineages (Doe, 2008 [↗](#)) and the cell-cycle blocked lineages have few or not progeny, we could not rule out the possibility that type 2 NSCs with cell cycle arrest failed to undergo normal temporal progression indirectly due to a lack of feedback signaling from their progeny. To address this concern, we generated random single NSC clones driving Cdk1RNAi, which developed alongside other wild-type type 2 NSCs. If the NSCs are still delayed in their temporal transition despite being in the same environment as wild-type NSC lineages, we can conclude that the cell cycle plays a role independent of its environment in the temporal progression of these NSCs. We found that, indeed, all type 2 NSC clones failed to switch from early Imp expression to EcR and Syp expression in late developmental stages (**Figure 2A-F** [↗](#)), demonstrating the intrinsic ‘clock’ that NSCs have in regulating their TTF progression independent of animal development. However, we cannot rule out the lineage-specific roles of progeny in providing feedback signals to regulate the timing of the temporal gene expression switch. During cortical development, both cell cycle and extrinsic factors govern neuronal fate, and it was proposed that neuronal progenitors undergo cyclic changes in their ability to respond to extrinsic cues (McConnell and Kaznowski, 1991 [↗](#); Okamoto et al., 2016 [↗](#)). Similarly, in *Drosophila*, Delta expression in the neighboring GMC and glia regulate Notch signaling, which is required for terminating neurogenesis in the central brain (Sood et al., 2024 [↗](#)). Further studies are needed to delineate the mechanism by which cell-cycle influences EcR expression and to confirm the role of progeny within the lineage in regulating temporal gene expression transitions within NSCs.

Materials and Methods

Immunohistochemistry

Larval brains were dissected in PBS and mounted on poly-D-lysine-coated coverslips. Samples were fixed for 23 minutes in 4% PFA in PBST, and then washed in PBST for 3×20 minutes. Following, the samples were blocked with 2% normal donkey serum and 2% normal goat serum (Jackson ImmunoResearch Laboratories, Inc.) in PBST for 40 minutes at room temperature. Samples were then incubated in a primary antibody mix diluted in PBST overnight or for 1-2 days at 4°C. Primary antibodies were removed, and samples were thoroughly washed with PBST. Samples were then incubated in secondary antibodies for 2 hours at room temperature. Secondary antibodies were removed, and samples were washed in PBST. Samples were dehydrated with an ethanol series of 30%, 50%, 75%, and 100% ethanol, then incubated in xylene (Fisher Chemical X5-1) for 2×10 minutes. Samples were mounted onto slides with DPX (Sigma-Aldrich 06552) and cured for 3-4 days, then stored at 4°C until imaged.

Image acquisition and analysis

Fluorescent images were acquired on both Zeiss LSM 710 and 800. Scale bars were given from all stacks within maximum intensity projection images. Brightness and contrast were adjusted in images uniformly for better visualization across the entire image across control and experimental samples. Statistics and bar graphs were computed using Prism 10.

Figure Preparation

Confocal images were prepared using FIJI or Imaris 10.0.0. Figures and schematics were assembled using Adobe Illustrator (2025).

Generation of clones

To generate RNAi clones, *hsFLP;10xUASmc-d8::GFP; ActFRTstopFRTGAL4* flies were crossed to the RNAi lines for the gene of interest or control *UAS-mCherryRNAi* line. Embryos were collected over a period of 4 hr. After hatching, larvae were collected and then heat shocked in a 37°C water bath for 8 minutes and reared at 25°C until the desired time point.

Supplementary figures

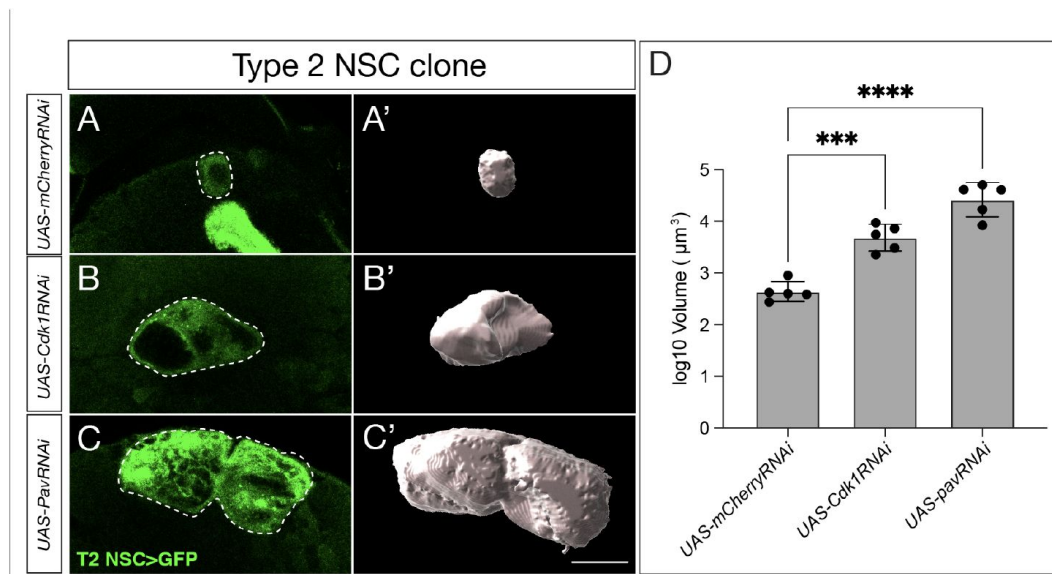


Figure S1

***pav* and *Cdk1*RNAi type 2 NSC clones are larger in volume.**

Confocal scans with their respective Imaris reconstruction are shown on the right for mCherryRNAi (A-A'), Cdk1RNAi (B-B'), and pavRNAi (C-C') clones. (D) Quantification of log¹⁰-transformed clone volumes (μm³); n = 5 clones per condition, plotted as mean ± SEM. Overall treatment effect by Welch's ANOVA, p < 0.0001. Post-hoc Dunnett's T3 vs. mCherryRNAi: Cdk1RNAi, p = 0.0003; pavRNAi, p < 0.0001. only one clone was quantified per animal. Scale bar represents 20μm.

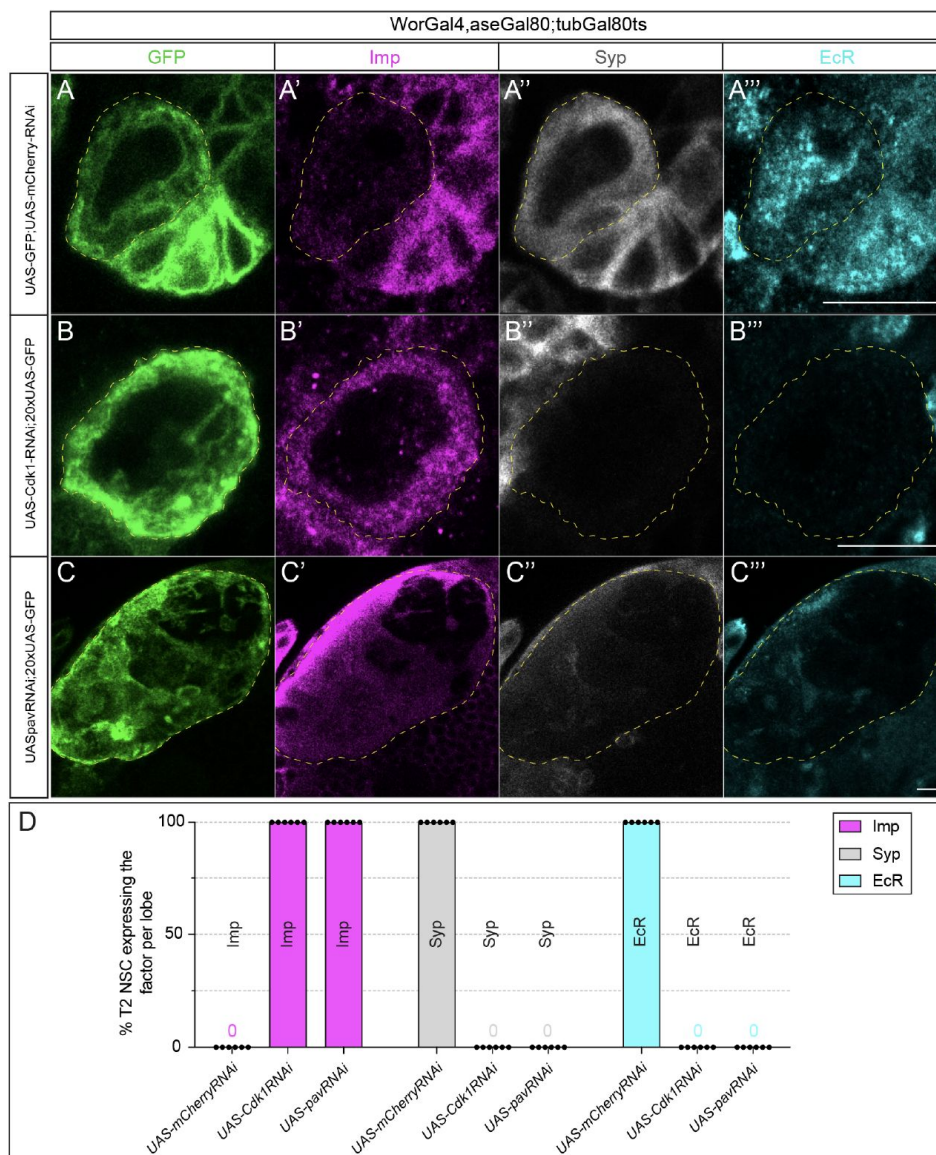


Figure S2

Cell-cycle and cytokinesis inhibit the early-to-late transition of temporal factors in type 2 NSCs.

(A-A''') Type 2 NSC control UAS-mCherry-RNAi (circled) at 72h ALH show normal temporal gene expression progression. Early factor Imp is off, and late factors Syp and EcR are on at 72h ALH (B-B'''). Cdk1RNAi type 2 NSCs fail to downregulate early factor Imp, and late factors Syp and EcR are not turned on. (C-C''') Similarly, cytokinesis-blocked type 2 NSCs using pavRNAi fail to downregulate Imp and up-regulate Syp and EcR. (D) Quantifications of early and late temporal factor expression in control, cell cycle, and cytokinesis blocked type 2 NSCs; n=6 for each genotype. Type 2 NSCs are identified as large cells expressing UAS-mcd8::GFP. Scale bars represent 10um.

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Additional information

Author contributions

Conceptualization: G.M.C and M.H.S.; methodology: G.M.C.; visualization: G.M.C. and M.H.S.; writing-original draft: G.M.C.; editing: M.H.S.; funding acquisition: M.H.S. and supervision: M.H.S.

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Author information

Gonzalo N Morales Chaya

Neural Diversity Lab, Department of Biology, University of New Mexico, Albuquerque, United States, Institute of Neuroscience, Howard Hughes Medical Institute, University of Oregon, Eugene, United States

ORCID iD: [0009-0005-4263-1323](https://orcid.org/0009-0005-4263-1323)

Mubarak Hussain Syed

Neural Diversity Lab, Department of Biology, University of New Mexico, Albuquerque, United States

ORCID iD: [0000-0003-2424-175X](https://orcid.org/0000-0003-2424-175X)

For correspondence: FlyGuy@unm.edu

Editors

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Reviewer #1 (Public review):

Summary:

Drosophila larval type II neuroblasts generate diverse types of neurons by sequentially expressing different temporal identity genes during development. Previous studies have shown that the transition from early temporal identity genes (such as *Chinmo* and *Imp*) to late temporal identity genes (such as *Syp* and *Broad*) depends on the activation of the expression of *EcR* by *Seven-up* (*Svp*) and progression through the G1/S transition of the cell cycle. In this study, Chaya and Syed examined whether the expression of *Syp* and *EcR* is regulated by cell cycle and cytokinesis by knocking down *CDK1* or *Pav*, respectively, throughout development or at specific developmental stages. They find that knocking down *CDK1* or *Pav* either in all type II neuroblasts throughout development or in single-type neuroblast clones after larval hatching consistently leads to failure to activate late temporal identity genes *Syp* and *EcR*. To determine whether the failure of the activation of *Syp* and *EcR* is due to impaired *Svp* expression, they also examined *Svp* expression using a *Svp-lacZ* reporter line. They find that *Svp* is expressed normally in *CDK1* RNAi neuroblasts. Further, knocking down *CDK1* or *Pav* after *Svp* activation still leads to loss of *Syp* and *EcR* expression. Finally, they also extended their analysis to type I neuroblasts. They find that knocking down *CDK1* or *Pav*, either at 0 hours or at 42 hours after larval hatching, also results in loss of *Syp* and *EcR* expression in type I neuroblasts. Based on these findings, the authors conclude that cell cycle and cytokinesis are required for the transition from early to late temporal identity genes in both types of neuroblasts. These findings add mechanistic details to our understanding of the temporal patterning of *Drosophila* larval neuroblasts.

Strengths:

The data presented in the paper are solid and largely support their conclusion. Images are of high quality. The manuscript is well-written and clear.

Weaknesses:

The quantifications of the expression of temporal identity genes and the interpretation of some of the data could be more rigorous.

(1) Expression of temporal identity genes may not be just positive or negative. Therefore, it would be more rigorous to quantify the expression of *Imp*, *Syp*, and *EcR* based on the staining intensity rather than simply counting the number of neuroblasts that are positive for these genes, which can be very subjective. Or the authors should define clearly what qualifies as "positive" (e.g., a staining intensity at least 2x background).

(2) The finding that inhibiting cytokinesis without affecting nuclear divisions by knocking down *Pav* leads to the loss of expression of *Syp* and *EcR* does not support their conclusion that nuclear division is also essential for the early-late gene expression switch in type II NSCs (at the bottom of the left column on page 5). No experiments were done to specifically block the nuclear division in this study. This conclusion should be revised.

(3) Knocking down *CDK1* in single random neuroblast clones does not make the *CDK1* knockdown neuroblast develop in the same environment (except still in the same brain) as wild-type neuroblast lineages. It does not help address the concern whether "type 2 NSCs with cell cycle arrest failed to undergo normal temporal progression is indirectly due to a lack of feedback signaling from their progeny", as discussed (from the bottom of the right column on page 9 to the top of the left column on page 10). The *CDK1* knockdown neuroblasts

do not divide to produce progeny and thus do not receive a feedback signal from their progeny as wild-type neuroblasts do. Therefore, it cannot be ruled out that the loss of Syp and EcR expression in CDK1 knockdown neuroblasts is due to the lack of the feedback signal from their progeny. This part of the discussion needs to be clarified.

(4) In Figure 2I, there is a clear EcR staining signal in the clone, which contradicts the quantification data in Figure 2J that EcR is absent in Pav RNAi neuroblasts. The authors should verify that the image and quantification data are consistent and correct.

<https://doi.org/10.7554/eLife.108259.1.sa2>

Reviewer #2 (Public review):

Summary:

Neural stem cells produce a wide variety of neurons during development. The regulatory mechanisms of neural diversity are based on the spatial and temporal patterning of neural stem cells. Although the molecular basis of spatial patterning is well-understood, the temporal patterning mechanism remains unclear. In this manuscript, the authors focused on the roles of cell cycle progression and cytokinesis in temporal patterning and found that both are involved in this process.

Strengths:

They conducted RNAi-mediated disruption on cell cycle progression and cytokinesis. As they expected, both disruptions affected temporal patterning in NSCs.

Weaknesses:

Although the authors showed clear results, they needed to provide additional data to support their conclusion sufficiently.

For example, they need to identify type II NSCs using molecular markers (Ase/Dpn).

The authors are encouraged to provide a more detailed explanation of each experiment. The current version of the manuscript is difficult for non-expert readers to understand.

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Reviewer #3 (Public review):

Summary:

The manuscript by Chaya and Syed focuses on understanding the link between cell cycle and temporal patterning in central brain type II neural stem cells (NSCs). To investigate this, the authors perturb the progression of the cell cycle by delaying the entry into M phase and preventing cytokinesis. Their results convincingly show that temporal factor expression requires progression of the cell cycle in both Type 1 and Type 2 NSCs in the *Drosophila* central brain. Overall, this study establishes an important link between the two timing mechanisms of neurogenesis.

Strengths:

The authors provide solid experimental evidence for the coupling of cell cycle and temporal factor progression in Type 2 NSCs. The quantified phenotype shows an all-or-none effect of cell cycle block on the emergence of subsequent temporal factors in the NSCs, strongly

suggesting that both nuclear division and cytokinesis are required for temporal progression. The authors also extend this phenotype to Type 1 NSCs in the central brain, providing a generalizable characterization of the relationship between cell cycle and temporal patterning.

Weaknesses:

One major weakness of the study is that the authors do not explore the mechanistic relationship between the cell cycle and temporal factor expression. Although their results are quite convincing, they do not provide an explanation as to why Cdk1 depletion affects Syp and EcR expression but not the onset of *svp*. This result suggests that at least a part of the temporal cascade in NSCs is cell-cycle independent, which isn't addressed or sufficiently discussed.

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